Gimbal Mount Assembly
Requirement Consolidation

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1. Introduction

1.1. Scope

Based on the preliminary requirements initially shared by TEC for this activity [RD1], this document refines the key performance parameters. These updated requirements primarily relate to the Hopper Ground Demonstrator *Oneiros*, which is also part of a de-risking activity for the *Huracan* propulsive system [RD2]. During the development of the GMA, the focus will be on prioritizing *Oneiros* requirements, while also considering intentions for upcoming design iterations of the *Huracan* engine and its components.

1.2. Reference

[RD1] Consultant Agreement 12.12.2025 - Docusign Envelope ID: AA5722EA-0E20-45A6-BFE2-2B6D313A7FF3

[RD2] Design Description and Justification File - TEC-ITA-DOC-2025-01011 - Version 3

[RD3] System Requirements Document - TEC-ITA-DOC-2025-01017 - Version 0

[RD4] Modern Engineering for design of liquid propelleant rocket engines - D.Huzel / D.Huang, Vol. 147, p. 341

2. Requirements

2.1. Environmental requirements

REQ-00X - Pollution requirements
Sensitive areas, such as regions of relative motion, shall be protected against sand particles with an average size of 0.2 mm at a temperature of 49.6 °C [RD3].

REQ-00X - Ambient requirements

According to the system requirements of *Oneiros*, the sea level conditions are defined by an pressure of 0.994 bar to 1.0276 bar and 12 % to 91 % humidity. The ambient temperature ranges from 8.3 °C to 49.6 °C. [RD3]

2.2. Functional requirements

REQ-003 - Operational gimbal capability
The GMA design provides a deflection capability of +/-10 ° per axis (pitch and yaw), which cover´s GNC needs and mechanical limits of the engine components (e.g. bellows, actuators).

REQ-00X - Max. gimbal capability
For off-nominal cases and emergency authority of the design, the max. gimbal angle shall not exceed +/-12 °.

REQ-002 - Rotation axis
To provide full pitch and yaw control capability, a two-axis rotation mechanism shall be implemented. Both axes are positioned perpendicular to each other within the same horizontal plane.

REQ-004 - Angular velocity
To compromise GNC needs with mechanical constraints of the engine such as the actuator capacity, the max. angular velocity of the engine is 20 °/s.

REQ-005 - Angular acceleration
The engine´s acceleration and hence, the capability of the GMA shall be max. 25 rad/s².

REQ-0014 - Mass
The maximum mass of the GMA shall be equal or below 5 kg with an mass measurement accuracy of +/-0.1 %.

REQ-00X - Geometrical envelope
A damage- and collision-free motion under worst case angle conditions is required between the Igniter´s main body (incl. it´s attachments) and the GMA geometry. A minimum clearance of 20 mm is to be considered between moving parts. The maximum geometrical limit of th GMA is given by the mass constraint, the loads and (thermo-)mechanical stress requirements mentioned in this document. The Ignition System´s main body (Ø70 mm) is designed with a vertically offset by 60 mm from the IH upper flange surface.

REQ-015 - Geometrical interface
For the lower attachment of the GMA, the usable surface of the IH features a ring with an inner diameter of Ø100 mm and an outer diameter of Ø170 mm. In total, eight through-holes (Ø9 mm each) are available, spaced at 45° intervals (symetrically along PCD), to fasten the GMA to the IH using eight M8 fasteners (minimum tensile ultimate strength per fastener: 800 MPa).

2.3. Mechanical and thermal requirements

REQ-018 - Friction coefficient
To reduce the breaking torque of the actuators, the GMA shall provide a constant friction coefficient equal or lower than 0.1 under mechanical and thermal operating conditions.

REQ-008 - Plastic Deformation and crack formation
The GMA and its parts will not undergo plastic deformation, crack formation or other non-recoverable deformation when subjected to the operational conditions defined in this specification. Exceptions can be permissible if non-recoverable deformation is identified as a single, local phenomena, that can be justified by analysis.

REQ-001 - Operating nominal thrust
The GMA shall transmit a total nominal thrust load of 15 kN.

REQ-00x - (Quasi-)static thrust
For single events, the GMA shall sustain an escessive load of 22.5 kN, which includes a safety factor of 1.5 applied to the nominal thrust.

REQ-030 - Cycling
The GMA shall sustain a minimum of 1000 cycles, which is derived from preliminary GNC-data.

REQ-020 - Operating temperature
The nominal operating temperature at the GMA surface that is connected to the IH flange lies at 120 K to 150 K.

REQ-021 - Thermal transient Until steady state conditions are achieved, transient conditions dominate. A cooling from ambient 280 K to 180 K within 10 s is to be considered linearly, with a slope of 10 K/s. This is to be taken into account at the GMA surface, that is connected to the IH upper flange.

2.4. Growth potential requirements

REQ-00x - Cycles

For further utilization of the GMA for flight qualification and preperation for the moon mission, a value of 26520 cycles is targeted as long as no inspection intervals are intended.

REQ-00x - Geometrical envelope
The GMA design must be flexible to adapt to different geometrical conditions. Future plans, where the Ignition System is intended to be printed together with the IH as one compact part, should be taken into account.

REQ-00x - Lateral misalignment
To reduce demands to vehicle´s guidance and the engine-actuation system, the thrust vector must be prealigned in all three axis. Hence, the GMA shall contain features to compensate lateral misalignment up to +/-10 mm. [RD4]

REQ-00x - Vacuum environment
The GMA performance in ambient conditions shall be adapted to operate feasibly in vacuum, especially concerning lubrication, thermal and consequently mechanical loads.

3. Acronym List

The acronyms used in this document are listed below.

Acronym	**Definition**
IH	Injection Head flange
GMA	Gimbal Mount Assembly
PCD	Pitch Circle diameter
QSL	Quasi-static load
TEC	The Exploration Company